Comparative Investigation on Tensile Performance of FRP Bars after Exposure to Water, Seawater, and Alkaline Solutions

Chunhua Lu[1]; Mingzhi Ni[2]; Tianshu Chu[3]; and Liyuan He[4]

Abstract: This paper presents experimental investigations to compare the effect of three environmental solutions on the degradation of tensile performance of fiber reinforced polymer (FRP) bars. Three kinds of FRP bars, basalt FRP (BFRP), carbon FRP (CFRP), and E-glass FRP (GFRP), were tested in tension after conditioning in tap water, artificial seawater, and alkaline solutions at room temperature for different exposure periods of 0, 45, 90, 135, and 180 days. The test results showed that the changes of the surface appearance of GFRP bars were more obvious that those of BFRP and CFRP bars after 180 days of exposure to three environmental solutions. The tensile failure mode and load-displacement curves of all tested FRP bars exhibited similar characters regardless of the immersion solution, the exposure period, or type of FRP bars. By comparing the residual mechanical properties of three kinds of FRP bars tested in this study, it has been found that the chemical attack of harsh environments simulated by immersion in seawater solution and alkaline solution resulted in a significant damage to BFRP bars. The effect of the seawater solution on the strength loss of CFRP bars was more obvious than those of the other two solutions. While the alkaline solution caused the greatest strength damage on GFRP bars. These findings were in accordance with the properties of material susceptibility to aggressive solutions and the moisture absorption of the matrix. No significant degradation of the elastic modulus of the three kinds of FRP bars had been found after exposure to aggressive solutions because their glass transition temperatures were kept unchanged in the experiment. Finally, based on the Arrhenius theory, the time-dependent tensile strength retention of all tested FRP bars exposed to the three aggressive solutions were fitted with experimental data obtained in this study. They were also compared with various test results collected from previous studies. The comparison results indicated that our experimental data were in good agreement with the predictions and most of the existing results. **DOI: 10.1061/(ASCE)MT.1943-5533.0003243.**

Author keywords: FRP bars; Aggressive solutions; Tensile strength; Elastic modulus; Ultimate strain; Prediction model.

Introduction

For the past several decades, the corrosion of steel bars in concrete has been considered one of the main factors of deterioration for reinforced concrete (RC) structures (Gjørv 2014; Lu et al. 2017b), especially when exposed to a chloride environment (Lu et al. 2017a). Several remedial measures, including replacing deteriorated reinforcements and using epoxy-coated or galvanized steel rebar, have been adopted in these deteriorated or possibly corroded RC structures (Sagüés et al. 2010). Recently, fiber reinforced polymer (FRP) bars have been believed to be another possible material to solve this problem (Robert and Benmokrane 2013; Ge et al. 2015; Ibrahim et al. 2016). They have been used in some concrete structures subjected to serious environmental conditions due to their advantage of chemical resistance (Micelli and Nanni 2004; Behnam and Eamon 2013). A high tensile strength might be another advantage for FRP bars with a similar elastic modulus to steel bars, such as CFRP bars. However, for GFRP bars with a low elastic modulus, the high tensile strength cannot be exploited in GFRP bars-reinforced concrete beams due to the limits or control requirements of deflection and crack width (D'Antino and Pisani 2018).

In addition, the doubt about their long-term performance (mainly, strength degradation and durability) in concrete is still an obstacle against their wide use in the world. Although it is commonly recognized that the FRP bars don't exhibit corrosion compared with steel bars, a variety of studies have confirmed that the tensile strength of FRP bars may result in a significant reduction when exposed to some typical conditions, like marine and alkaline environments (Micelli and Nanni 2004; Silva et al. 2014; Hwang et al. 2014). It was widely reported that the corrosion of steel bars in concrete usually gives direct and visible warning signs, such as cracking or spalling of concrete (Shi et al. 2017). However, the deterioration of FRP bars in concrete does not reflect these visible phenomena through the cover (Al-Salloum et al. 2013; Behnam and Eamon 2013). Although it seems beneficial, no warning sign existing is a deficiency for the FRP-reinforced concrete members because of their brittle failure characteristic (Al-Salloum et al. 2013). As a result, the degradation of chemical resistance of FRP bars exposed to different kinds of aggressive solutions, such as water, saline, and alkaline solutions, is required to be effectively evaluated and characterized (Lu et al. 2016).

Table 1 summarizes the test solutions used to evaluate the mechanical and durability performance of FRP bars. It can be observed from Table 1 that the pH value of an alkaline solution is

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Table 1. Aggressive solutions designed to evaluate the performance of FRP bars

Reference	Solution	Quantities in g/L								pH
		$Ca(OH)_2$	NaOH	KOH	NaCl	Na_2SO_4	$MgCl_2$	$CaCl_2$	KCl	
ACI Committee 440 (2004)	Alkaline solution	118.5	0.9	4.2	—	—	—	—	—	12.9
ASTM (2014c)	Seawater solution	—	—	—	24.54	4.09	5.2	1.16	0.69	7.1
Micelli and Nanni (2004)	Alkaline solution	1.6	10	14	—	—	—	—	—	13.0
Chen et al. (2006, 2007)	Tap water	—	—	—	—	—	—	—	—	—
	Alkaline solution 1	2	2.4	16.9	—	—	—	—	—	13.06
	Alkaline solution 2	0.037	0.6	1.4	—	—	—	—	—	12.7
	Ocean water	—	—	—	30	5	—	—	—	7.0
	Combined solution	—	—	5.6	71.66	—	—	—	—	13.0
Kim et al. (2008)	Tap water	—	—	—	—	—	—	—	—	7.0
	NaCl solution	—	—	—	30	—	—	—	—	7.0
	Alkaline solution	1.6	10	14	—	—	—	—	—	13.0
Won et al. (2008)	Alkaline solution	1.6	10	14	—	—	—	—	—	13.0
Al-Salloum et al. (2013)	Tap water	—	—	—	—	—	—	—	—	7.0
	Sea water	Solution taken from the Arabian Gulf—Eastern province								8.0
Wu et al. (2015)	Alkaline solution	2	0.9	4.2	—	—	—	—	—	12.9
Serbescu et al. (2015)	Water	—	—	—	—	—	—	—	—	7.0
	Alkaline solution	118.5	0.9	4.2	—	—	—	—	—	13.0
Micelli et al. (2017)	Alkaline solution	1.6	—	—	—	—	—	—	—	12.6
Micelli and Aiello (2019)	Environment A	1.6	—	—	—	—	—	—	—	12.6
	Environment B	1.6	21	14	—	—	—	—	—	13.0
	Environment C	—	—	2	—	—	—	—	—	12.5
	Environment D	—	50	—	—	—	—	—	—	14.0

usually around 13.0 [ACI 440 (ACI 2004); Chen et al. 2006; Kim et al. 2008; Won et al. 2008; Wu et al. 2015; Serbescu et al. 2015; Micelli et al. 2017; Micelli and Aiello 2019], although its composition of alkali hydroxide is diverse. The saline solution is generally prepared with the simulated seawater solution (ASTM 2014c), NaCl solution (Chen et al. 2006; Kim et al. 2008), or local sea water (Al-Salloum et al. 2013) in which the pH value is approximate 7.0. Under the environmental conditions of these aggressive solutions, many experimental works have been conducted in recent decades to explore the mechanical performance of FRP bars with various exposure periods and temperatures. For the E-glass FRP (GFRP) bars, some researchers had completed the solution immersion test with elevated temperatures and found that alkaline solution a had more harmful effect on the tensile strength than water and saline solutions (Kim et al. 2008; Al-Salloum et al. 2013). It has also been found that degradation was accelerated through the use of elevated temperatures for the same exposure condition (Chen et al. 2006; Kim et al. 2008; Won et al. 2008; Al-Salloum et al. 2013). Unlike the well-developed GFRP bars, the degradation of the mechanical and the durability of the performance of BFRP bars remains unclear when they are exposed to aggressive environmental conditions. Wu et al. (2015) evaluated the residual tensile properties of BFRP bars exposed to an alkaline solution, salt solution, acid solution, and deionized water at 25°C, 40°C, and 55°C and found that the durability of BFRP bars was less influenced by the salt, acid, and water solutions compared with the alkali solution. Furthermore, based on a long-term performance prediction with the Arrhenius theory, they calculated that for the 6 mm BFRP bars, the exposure time in an alkaline solution at a mean annual temperature of 5.7°C, which represents an area with a northern latitude of 50°, was estimated at approximately 16.1 years when a reduction in strength of 50% was achieved. Serbescu et al. (2015) conducted the accelerating experiment on two types of BFRP bars immersed in two kinds of solutions (Table 1) at 20°C, 40°C, and 60°C for a total period of 5,000 h and estimated from the results that the tensile strength of BFRP bars would retain about 72% after 100 years' exposure to a concrete environment. By comparing the predictions obtained in two preceding research works, it can be concluded that the obviously different findings have been proposed due to the various discrepancies in BFRP material and exposure conditions and that the corresponding further research works are necessarily required.

Besides, some researchers have carried out the experiments to compare the durability performance of two types of FRP bars directly. Wang et al. (2017) conducted accelerated corrosion tests to explore the long-term performance of BFRP and GFRP bars in two types of seawater and sea sand concrete environments and proposed, from the tensile experiments, that the BFRP bars seemed to be more durable than GFRP bars, while the elastic modulus of all specimens remained not significantly changed. Chen et al. (2007) experimentally compared the durability of CFRP and GFRP bars immersed in the simulated pore solution of normal concrete (alkaline solution 1 in Table 1) at 60°C for 70 days and found than CFRP bars displayed more excellent durability performance than GFRP bars. It seems that the BFRP and CFRP bars have a better ability to chemical resistance than GFRP bars. However, Micelli and Aiello (2019) concluded from experiments that E-glass fibers and basalt fibers confirmed to be highly sensitive to alkaline solutions, while carbon fibers did not exhibit a chemical vulnerability in alkaline solutions.

From the previous introductions, it can be concluded that many researchers have investigated the durability performance of FRP bars exposed to different aggressive solutions. However, it should be noted that the preceding research had many discrepancies compared with the newest reported tensile test results. It might be attributed to the differences of experimental environments and the FRP bars adopted. Besides, in order to get accelerated aging, the aforementioned studies paid more attention to the combined effect of an aggressive solution and elevated temperature on the degradation of mechanical properties of FRP bars. However, from the practical application in civil engineering, it is imperative and valuable to investigate the tensile performance of FRP bars in a normal civil engineering environment, especially under natural exposure temperatures (D'Antino et al. 2018).

This paper aims to comparatively investigate the effect of three natural aggressive environmental solutions on the degradation of tensile performance of BFRP, CFRP, and GFRP bars. Firstly, the natural immersion experiments of these three kinds of FRP bars were carried out in alkaline, artificial seawater, and tap water solutions with a total exposure period of 180 days. The moisture absorption test and a differential scanning calorimetry (DSC) analysis were conducted during the immersion tests. Then, the tensile properties of all unconditioned and conditioned FRP bars were measured after exposure of 0, 45, 90, 135, and 180 days. Based on test results, the tensile strength retentions of three types of FRP bars were compared, and the effects of the aggressive solutions and exposure periods on the degradation of tensile properties were discussed. Finally, the prediction model of the tensile-strength retention based on the Arrhenius theory was applied to fit the experimental data obtained in this study, and the results were widely compared with the outcomes reported in previous studies.

Experimental Program

Materials

Three types of FRP bars, the BFRP bar, CFRP bar, and GFRP bar, with a nominal diameter of 8 mm were prepared for the experiment, and all of them were industrially manufactured by Nanjing Fenghui Composite Material Co. Ltd. located in Jiangsu Province of China. BFRP, CFRP, and GFRP bars tested in this study were made of a vinyl ester matrix and corresponding fibers (basalt fibers, carbon fibers, and E-glass fibers, respectively) with a volume fraction of fibers equal to 64%. The original tensile strength and elastic elasticity of vinyl ester matrix provided by the manufacturer were 84.2 MPa and 3.4 GPa, respectively. Therefore, the main difference between them is the type of fiber they used. As presented in Fig. 1, the surfaces of the BFRP bar and GFRP bar were grooved during pultrusion, while the CFRP bar was helically wrapped at its surface. As a result, these treatments caused a slight effect on the diameter distribution along the bar length. The measured average outer diameter and inner diameter of BFRP and GFRP bars were 8.0 ± 0.1 mm and 7.9 ± 0.1 mm, respectively. For CFRP bars, the measured outer and inner diameters were rather close with the mean of 8.0 ± 0.1 mm. In this study, the nominal diameter was adopted when the cross-section area of FRP bars was calculated. The initial mechanical properties of these unconditioned bars tested in this study are shown in Table 2.

Aggressive Solution Environments

In this study, three solutions including tap water, artificial seawater, and alkaline solutions were prepared for FRP bars' immersion tests. The tap water solution was designed with ordinary tap water used in Zhenjiang, Jiangsu Province of China. The artificial seawater solution and alkaline solution were designed according to the test methods proposed by ACI 440.3R-04 (ACI 2004) and ASTM D665-14e1 (ASTM 2014c), respectively, as shown in Table 1. As a result, the measured pH values of the seawater solution and alkaline solution were 7.1 and 12.9, respectively.

Fig. 1. Three types of FRP bars used in this study.

Table 2. Mechanical properties of the FRP bars

	Tensile strength, f_{u0}		Modulus of elasticity, E_{f0}		Ultimate strain, ε_{f0}	
Bar type	Mean (MPa)	COV (%)	Mean (GPa)	COV (%)	Mean	COV (%)
BFRP	1,299.6	2.9	42.1	0.6	0.0309	2.3
CFRP	2,079.0	2.2	134.9	0.9	0.0154	3.1
GFRP	1,409.5	1.6	50.0	2.9	0.0282	1.3

Note: For each type of FRP bar, three tensile specimens were tested.

Before soaking, both ends with a length of 300 mm of the FRP sample were sealed with epoxy resin in order to avoid chemical attack at the two ends. The type of epoxy resin was WSR618 (E-51), which was produced by Bluestar New Chemical Materials Co. Ltd (Wuxi, China). Then, the FRP samples were immersed in different solution containers, which were placed in the authors' material lab, located in Zhenjiang, Jiangsu Province of China, with an average room temperature of $20°C \pm 2°C$. The evaporation of water in the solution container was considered by observing the amount of the solution, and their pH values were measured every ten days. It was found that the pH of the alkaline solution had some changes during exposure periods. When its pH value changed out of the range of 12.9 ± 0.2, the adjustment of the pH was completed by adding a certain amount of $Ca(OH)_2$ powder based on the measured pH value and solution volume. In this study, the pH value of the alkaline solution was adjusted after 60 and 120 days of exposure. The total duration of exposure was 4,320 h (180 days).

Sample Preparation for Tensile Test

For all unconditioned and conditioned FRP bars, the tensile samples should be prepared before the tensile test. In this study, the tensile test sample of FRP bars was designed with adhesive anchors at the two ends within the length of 300 mm, as shown in Fig. 2(a). The anchor used in this study was made of galvanized steel pipe with an inner diameter of 9 mm and thickness of 4 mm. Aluminum caps were settled to keep the bar in the center of the pipe during anchor installation. In order to ensure the adhesive force between the FRP bar and the anchor was large enough, the same epoxy resin of WSR618 (E-51) was used between them, and the mass ratio of the epoxy resin and the curing agent was 4:1. The total length of the tensile specimen was 1,000 mm, and some typical specimens ready for the tensile test were given in Fig. 2(b). The prepared tensile samples would be dried at 20°C for 1 week before the tensile test.

In order to obtain the quantitative description of the time-dependent degradation of FRP bars, tensile properties were investigated after exposure of 0, 45, 90, 135, and 180 days, respectively.

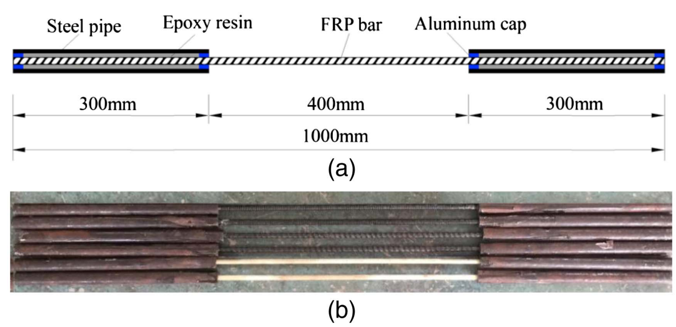

Fig. 2. Tensile specimens of FRP bar: (a) schematic diagram; and (b) typical specimens ready for tensile test.

For unconditioned bars with an exposure of zero days, their tensile properties were measured just before the immersion test, and the results are given in Table 2. While, for conditioned FRP bars, a number of specimens were taken out from the solution containers after the required exposure period. Then, they were cleaned to make the tensile samples. After the final exposure of 180 days, the surface appearance of the bars was observed and examined before cleaning. For each kind of FRP bar, three specimens immersed in a certain solution were tested at one time and the mean of the three results was taken as the test value. Thus, a total of 108 tensile specimens of conditioned FRP bars were investigated and compared in this study.

Tensile Tests

According to the guide described in section "B.2—Test method for longitudinal tensile properties of FRP bars" in ACI 440.3R-04 (ACI 2004), the tensile tests of all FRP bars were conducted using a CSS-44300 electrohydraulic universal testing machine (UTM) (Fig. 3).

Fig. 3. Setup of tensile test.

The loading process was controlled by displacement with a low rate of 2 mm/min (Li et al. 2015). Using its own clip-on extensometer, the tensile strain of the FRP bar was measured. All test data can be collected by the data acquisition system automatically. The data obtained were utilized to plot the load-displacement (or stress-strain) curve and determine the ultimate tensile strength and elastic modulus of all tested FRP bars.

Absorption Test

During the immersion test, the moisture absorption of conditioned FRP bars was assessed after exposure of 5, 10, 20, 40, 80, and 120 days. Before conditioning, the bar original weight was measured after being oven dried for 10 h at 40°C. After the exposure of the required time, the FRP specimens were removed from the solution containers, and their surface was wiped dry with filter paper. The mass of conditioned FRP bars was weighed immediately using an electronic balance with an accuracy of 0.001 g. After that, the specimens were put back into the containers. Finally, the moisture uptake of a FRP bar was calculated as the difference between the current and original measurements.

Differential Scanning Calorimetry

Using an STA 499F3 calorimeter, a differential scanning calorimetry (DSC) analysis was conducted to determine the glass transition temperature (T_g) values of the unconditioned and conditioned FRP bars. The FRP specimens with a weight of 10 mg from the tensile samples were heated from 30°C to 150°C at a rate of 2.5°C/min in a nitrogen environment. The glass transition temperature, T_g, was determined in accordance with the ASTM E1356-2008 standard (ASTM 2014a).

Results and Discussions

Surface Observation

For the FRP bars immersed in aggressive solutions for 180 days, their surface appearances were observed before cleaning, as shown in Fig. 4. It can be found from Fig. 4 that the appearance changes of GFRP bars were more obvious than those of BFRP and CFRP bars and some damage characteristics, such as resin spalling and surface deterioration with blisters or particles, appeared on the surface of the GFRP bar. For BFRP bars, some chemical attack phenomena caused by the alkaline solution could be seen on the bar surface and no obvious change appeared on the surface of the bars immersed in the seawater and tap water solutions. These findings showed that

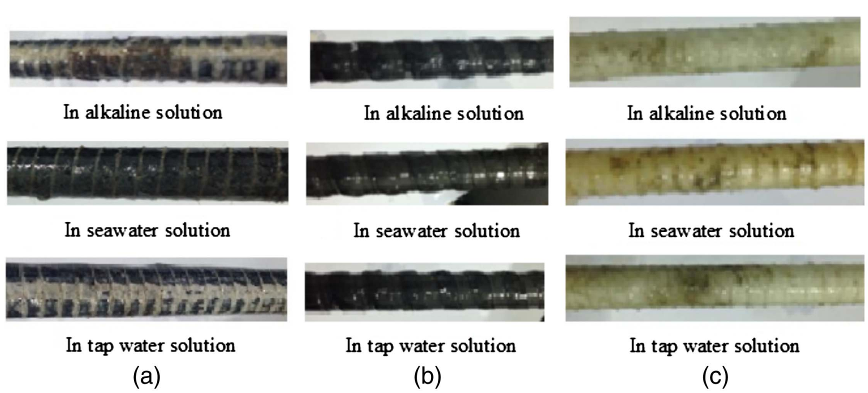

Fig. 4. Surface appearance of FRP bars after 180 days of immersion in aggressive solutions: (a) BFRP bar; (b) CFRP bar; and (c) GFRP bar.

the three types of FRP bars made with the same matrix presented the different surface characteristics after exposure to the same environment. The possible reasons for this phenomenon were the different surface treatments and moisture absorption contents of tested FRP bars. It seems that the helically wrapped CFRP bars have less change in surface properties after solution exposure. By comparing, it can be preliminarily concluded that the ability of the chemical resistance of CFRP bars is best among the three kinds of FRP bars tested in this study, as their surfaces remain intact in three aggressive solutions in this experiment.

Moisture Absorption

Based on the standard of ASTM D5229/D5229M-14e1 (ASTM 2014b), the average moisture content, M (%), in a FRP specimen was taken as the ratio of the average mass of moisture uptake to the original mass of an oven-dry specimen. Fig. 5 shows the moisture absorption expressed with M of conditioned FRP bars with respect to the exposure period up to 120 days. In Fig. 5, with increasing the exposure period, all FRP specimens showed gradual weight gain up to about 80 days. During this stage, the diffusion process of moisture in FRP material could be expressed with Fick's 2nd law. After that, the diffusion mechanism started to become unstable due to the degradation of the outer layer of the FRP specimen (Fergani et al. 2018).

For the three types of FRP bars, the results given in Fig. 5 indicated that the seawater solution caused greater moisture absorption than the other two solutions, while the moisture uptake in the alkaline solution was lowest. In addition, after the same exposure to seawater and tap water solutions, the moisture content in GFRP bars was higher than those in BFRP and CFRP bars. While after the same exposure to the alkaline solution, the moisture content in all FRP bars seemed to be equivalent.

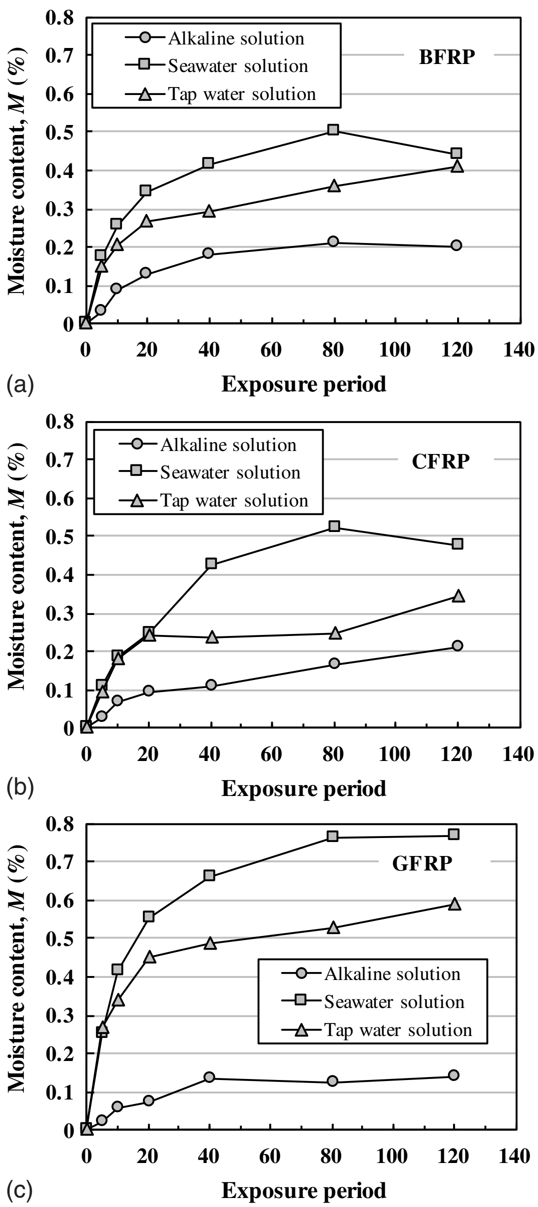

Fig. 5. Moisture absorption of tested FRP bars immersed in aggressive solutions: (a) BFRP bar; (b) CFRP bar; and (c) GFRP bar.

DSC Analysis

Based on the properties of surface change and moisture absorption of the aged FRP bars, the BFRP and CFRP bars immersed in seawater solution and the GFRP bars exposed to alkaline solution were selected to determine the T_g after exposure of 180 days, which were compared with the T_g values of unconditioned FRP bars. Fig. 6 presents the variation of the DSC heat flow first-derivative of tested FRP bars with an increasing temperature up to 150°C in a nitrogen environment. Based on the standard of ASTM E1356-2008 (ASTM 2014a), the T_g values of unconditioned and conditioned BFRP samples in the seawater solution were determined to be 59.7°C and 56.7°C, respectively. For CFRP bars, the corresponding T_g values were 58.3°C and 57.1°C. In addition, the T_g values of unconditioned and aged GFRP bars in alkaline solution were found to be 61.1°C and 60.7°C, respectively. These results of the DSC analysis showed that there were no significant changes of the T_g value in this experiment, which indicated no major effect of exposure solutions on the thermal properties of the vinyl ester matrix. The similar result was also found for GFRP bars embedded in moist concrete (Robert et al. 2009). Therefore, the conditioned FRP bars at room temperature can still maintain the original stiffness after exposure to aggressive solutions for 180 days.

Failure Mode and Load-Displacement Curve

In the tensile test, most of the unconditioned and conditioned FRP bars showed a similar failure mode in which the bar failed through

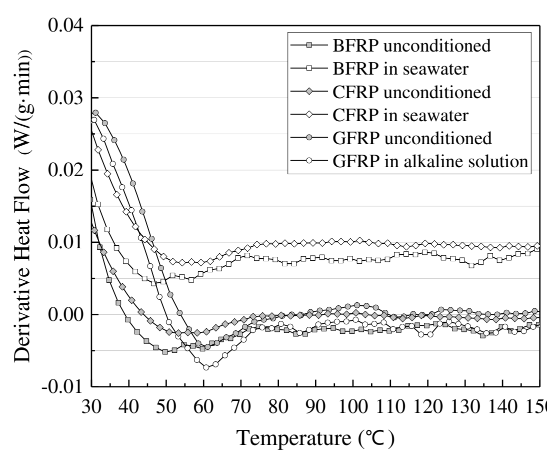

Fig. 6. DSC heat flow fist derivative-temperature curves of tested FRP bars.

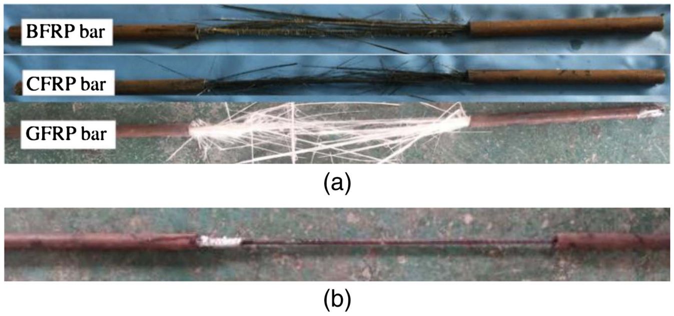

Fig. 7. Typical failure modes of all FRP bars: (a) typical failure mode; and (b) failure due to bar's slippage.

fiber rupture. The failure was accompanied by longitudinal delamination caused by the debonding of the fiber/matrix interface, as shown in Fig. 7(a). This mode of failure was a typical mode of brittle fracture, which usually occurred from the middle to the two ends of the bar, and the direction of delamination was parallel to the fibers. Comparing the failure of the three kinds of FRP bars, it seemed that the dispersion of fractured fibers of the GFRP bar was more obvious than those of the BFRP and CFRP bars. A similar tensile failure mode was also found for BFRP bars (Shi et al. 2017; Serbescu et al. 2015) and GFRP bars (Robert and Benmokrane 2013; Chen et al. 2006; Al-Salloum et al. 2013). It can be believed that there is no obvious effect of the aggressive solution and exposure period on the tensile failure mode of FRP bars. Besides, as the effective bond between the FRP bar and the anchor failed, only a few tensile tests were stopped owing to bar's slippage from the anchor [Fig. 7(b)], and the corresponding testing results were not considered in this study. This pull-out failure due to the significant slip between the bar and anchor was also reported by Serbescu et al. (2015).

Taking the environmental effect and the typical failure mode into account, Fig. 8 shows the representative load-displacement curves of the BFRP and CFRP bars immersed in the seawater solution and GFRP bars exposed to the alkaline solution. It should be noted that three specimens of one kind of FRP bar were tested at each time and only one load-displacement curve was selected to be plotted in Fig. 8. Besides, the similar curves could be obtained for the FRP bars immersed in other solutions. The load-displacement relationships of all FRP bars showed almost linear up to the tensile failure, regardless of the difference in environmental solutions and exposure periods, which indicated that the tensile properties of all tested FRP bars presented the approximately linear elastic behaviors no matter whether the bars were conditioned or not.

Tensile Strength

According to the tensile tests, the FRP bars failed suddenly when the applied load reached to the peak load, as shown in Fig. 8. The corresponding ultimate tensile strength of unconditioned and conditioned FRP bars can be evaluated based on Eq. (1) [ACI 440 (ACI 2004)]

$$f_u = \frac{P_u}{A_f} \tag{1}$$

where f_u = nominal ultimate tensile strength of the FRP bars (MPa), and the terms of f_{u0} and $f_{u,c}$ are used to express the tensile strengths of unconditioned and conditioned bars, respectively; P_u = tensile capacity (N); and A_f = cross-sectional area of the FRP bar. In this study, the diameter of the FRP bar, d, is assumed to be

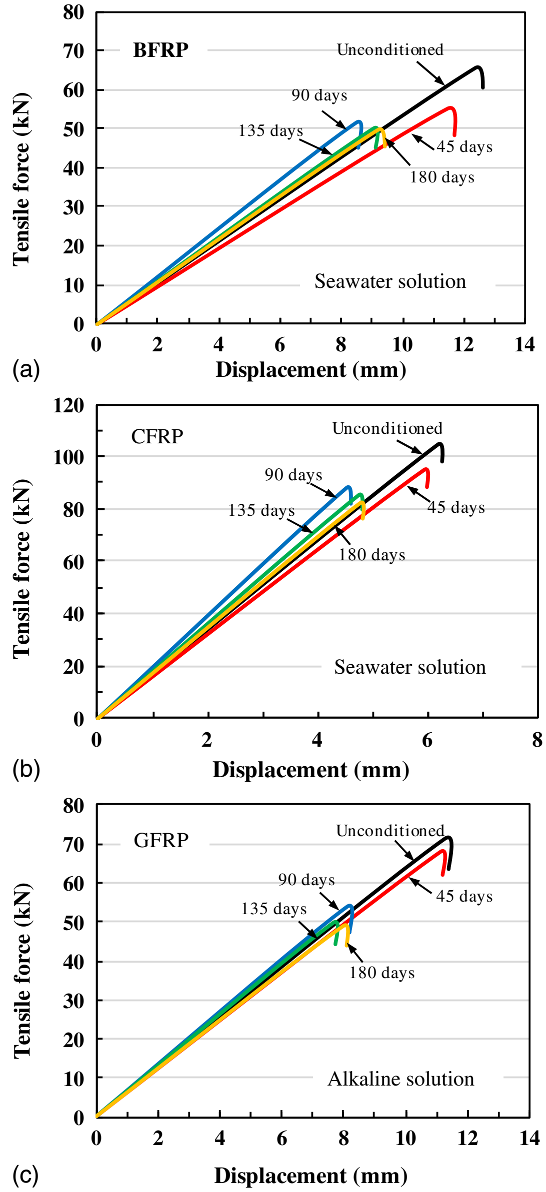

Fig. 8. Load-displacement curves of part FRP bars: (a) BFRP bars in seawater solution; (b) CFRP bars in seawater solution; and (c) GFRP bars in alkaline solution.

unchanged during the tensile test, and its original nominal diameter ($d = 8.0$ mm) is used to calculate the cross-sectional area.

Based on the curves shown in Fig. 8, the relationship between the applied load and bar strain calculated with the displacement and original length can be assumed to be linear up to the failure. Thus, the elastic modulus of FRP bars in this test can be determined using Eq. (2) [ACI 440 (ACI 2004)]

$$E_f = \frac{P_1 - P_2}{(\varepsilon_1 - \varepsilon_2)A_f} \tag{2}$$

where E_f = axial modulus of elasticity of FRP bars (GPa), and the terms of E_{f0} and $E_{f,c}$ stand for the elastic moduli of unconditioned and conditioned bars, respectively; P_1 and ε_1 = load (N) and corresponding strain, respectively, at approximately 50% of the ultimate tensile capacity; P_2 and ε_2 = load (N) and corresponding strain, respectively, at approximately 20% of the ultimate tensile capacity.

Table 3 collects the calculated results of the average tensile strength, elastic modulus, and ultimate strain of all conditioned FRP bars. It can be observed from Tables 2 and 3 that the tensile strengths of FRP bars immersed in different solutions at room temperature clearly degenerated with an increasing exposure period. In order to reflect the time-dependent degradation effect of environmental solutions on the tensile strength of FRP bars, a function of tensile-strength retention related to the exposure period, $\eta(t)$, was defined as the ratio of the bar tensile strength after exposure to its initial tensile strength. The corresponding expression can be equally shown in Eq. (3)

$$f_{u,c} = \eta(t) \cdot f_{u0} \tag{3}$$

Based on the results of the tensile strength given in Tables 2 and 3, the values of tensile strength retention $\eta(t)$ were obtained for all conditioned FRP bars and the corresponding calculations were plotted in Fig. 9. Comparing the outcomes shown in Fig. 9, several findings can be made:

Table 3. Results of tensile test of all conditioned FRP bars

Bar type	Environmental solution	Exposure period (Day)	Tensile strength Mean (MPa)	Tensile strength COV (%)	Elastic modulus Mean (GPa)	Elastic modulus COV (%)	Ultimate strain Mean (%)	Ultimate strain COV (%)
BFRP	Alkaline solution	45	1,194.7	2.3	44.3	1.8	0.0270	2.7
		90	1,148.9	2.6	46.2	2.1	0.0249	1.5
		135	1,078.5	4.2	43.9	2.8	0.0245	3.1
		180	1,008.6	5.1	44.7	2.3	0.0226	2.2
	Seawater solution	45	1,095.2	3.4	38.1	3.7	0.0287	2.4
		90	1,028.5	5.9	48.5	4.7	0.0212	1.8
		135[a]	998.7	7.2	44.1	4.5	0.0226	3.6
		180	984.8	6.4	43.0	1.6	0.0229	2.1
	Tap water solution	45	1,256.7	3.0	42.8	1.2	0.0293	1.8
		90	1,207.3	5.3	45.1	2.5	0.0268	3.3
		135	1,155.3	4.2	42.0	0.9	0.0276	2.7
		180	1,150.1	5.1	47.1	3.8	0.0245	4.6
CFRP	Alkaline solution	45	2,059.0	3.4	135.4	2.1	0.0152	2.2
		90	1,966.6	1.8	144.5	3.3	0.0136	3.5
		135	1,928.8	1.9	125.1	2.8	0.0154	2.9
		180	1,720.5	4.3	138.4	1.1	0.0125	2.6
	Seawater solution	45	1894.9	8.2	129.2	6.6	0.0147	5.3
		90[a]	1,758.7	7.5	157.1	7.2	0.0112	6.1
		135	1,692.5	5.6	142.7	3.9	0.0118	2.9
		180	1,638.3	6.9	137.9	3.4	0.0119	3.6
	Tap water solution	45	2,010.8	3.3	138.7	1.2	0.0146	2.1
		90	1,916.8	4.7	129.3	1.5	0.0148	3.2
		135[a]	1,879.4	5.9	130.9	2.3	0.0143	2.8
		180	1,848.2	4.6	140.8	1.6	0.0132	2.2
GFRP	Alkaline solution	45	1,359.8	2.1	48.9	3.3	0.0278	2.4
		90	1,061.4	4.2	52.3	3.5	0.0203	1.6
		135	994.7	3.4	51.8	2.2	0.0192	3.1
		180	974.8	1.8	49.1	1.7	0.0199	2.7
	Seawater solution	45	1,402.6	1.3	49.9	1.2	0.0281	3.2
		90	1,298.1	0.9	52.0	3.2	0.0250	3.4
		135	1,275.2	1.6	48.9	2.5	0.0261	2.0
		180	1,152.9	2.2	51.5	2.8	0.0224	1.8
	Tap water solution	45	1,372.0	1.5	51.9	2.3	0.0264	1.5
		90	1,264.4	1.9	50.7	2.1	0.0249	2.3
		135	1,202.7	3.5	50.2	2.6	0.024	2.6
		180	1,098.9	2.7	50.7	1.5	0.0217	1.9

[a]Only two effective data were obtained in the tensile test.

- For the three types of FRP bars tested in this study, it seems that the CFRP bars have better chemical resistance ability than BFRP and GFRP bars, and the GFRP bars display the worst durability performance.
- Considering the three environmental solutions used in this study, the chemical attack of harsh environments simulated by immersion in the seawater solution and alkaline solution resulted in a significant damage of BFRP bars. The effect of the seawater solution on the strength loss of CFRP bars was more obvious than those of the other two solutions. The alkaline solution caused greater damage on GFRP bars than the other two solutions.
- Taking the types of FRP bars and aggressive solutions into account, it can be indicated on the whole that the degradation of tensile strength of GFRP bars exposed to the alkaline solution is the most obvious, while a tap water solution causes the slightest damage on BFRP and CFRP bars.

These preceding phenomena can be explained with the coupled effects of material susceptibility to aggressive solutions and moisture uptake during the immersion periods. It is known that vinyl ester matrix is prone to degradation by hydrolysis in a water environment (Chen et al. 2007; Kim et al. 2008; Wang et al. 2017;

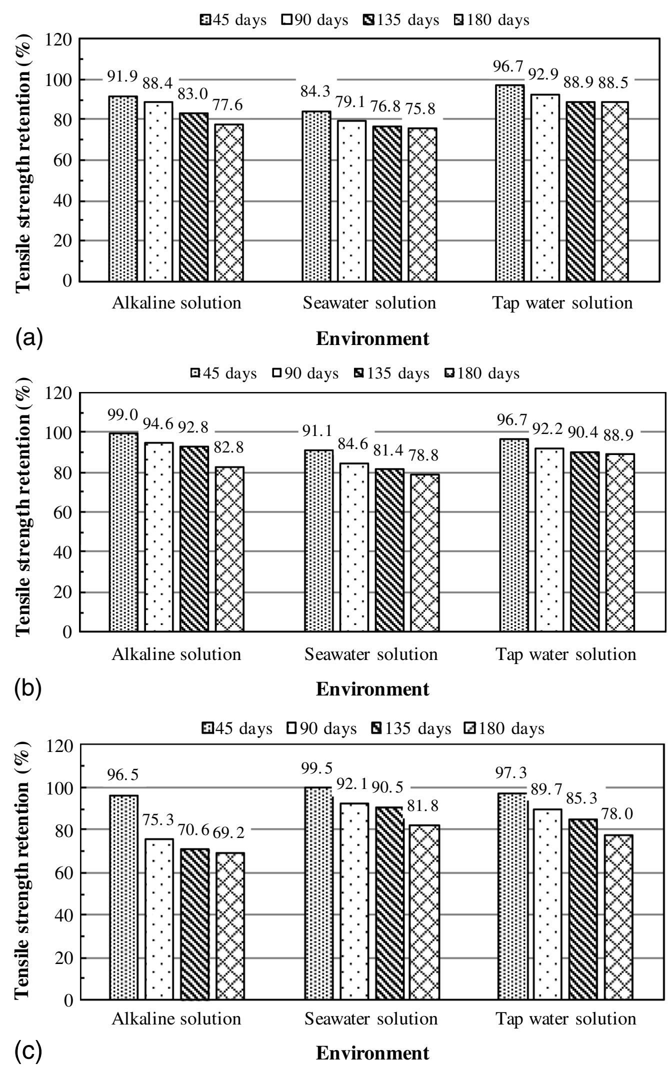

Fig. 9. Tensile strength retention of all conditioned FRP bars: (a) BFRP bars; (b) CFRP bars; and (c) GFRP bars.

Fergani et al. 2018). Therefore, the degradation of the matrix with more moisture uptake will become greater. From the results of the moisture content of tested FRP bars, as shown in Fig. 5, it can be observed that BFRP and CFRP specimens exposed to seawater solution showed the greatest moisture uptake among the three aggressive solutions. As a result, a seawater solution would cause a greater deterioration effect on the vinyl ester matrix in BFRP and CFRP bars than alkaline and tap water solutions, resulting in a greater degradation of tensile strength of BFRP and CFRP bars. For GFRP bars, except for the deterioration effect of water on the matrix, it is also known that the E-glass fibers are themselves extremely susceptible to degradation in the water and alkaline (hydroxyl ions) environments due to leaching and etching actions (Chen et al. 2007; Kim et al. 2008; Wang et al. 2017; Fergani et al. 2018). Therefore, the GFRP bars tested in this study are believed to be most sensitive to an alkaline solution. Therefore, the result that the alkaline solution caused the greatest damage on GFRP bars can be attributed to the susceptibility of GFRP bars to the alkaline solution.

Elastic Modulus

Based on the obtained results listed in Tables 2 and 3, Fig. 10 compares the tensile modulus of elasticity of all tested FRP bars with an increasing exposure period. From Fig. 10, it can be indicated that the elastic modulus of conditioned FRP bars mainly fluctuated in the range of 0.95–1.1 times the initial modulus of elasticity of unconditioned specimens. In Fig. 10(a), the elastic modulus of conditioned BFRP specimens slightly increased (less than 10%) with an increasing exposure period. For BFRP and CFRP bars, it can be noticed from Figs. 10(a and b) that the seawater solution caused a greater fluctuation of the elastic modulus than the other two solutions. By comparison, the effect of aggressive solutions on the modulus of elasticity of GFRP bars seemed to be slightest among the tested three kinds of FRP bars.

In general, no significant degradation of elastic modulus of three kinds of FRP bars had been found after exposure to alkaline, seawater, and tap water solutions tested in this study. This result might be attributed to the fact that the glass transition temperature of tested FRP bars hadn't been changed after exposure to aggressive solutions at room temperature. Similar observations of the elastic modulus of FRP bars exposed to various environmental conditions were also found for BFRP bars (Serbescu et al. 2015; Wang et al. 2017, 2018) and GFRP bars (Robert and Benmokrane 2013; Al-Salloum et al. 2013; Wang et al. 2017; Nkurunziza et al. 2005; Robert et al. 2009; Wang et al. 2018).

Ultimate Strain

Fig. 11 displays the obtained ultimate strains of all tested FRP bars exposed to different aggressive solutions. Due to the fact that the modulus of elasticity of conditioned FRP bars remained approximatively unchanged in the aggressive solutions, it can be observed from Fig. 11 that the ultimate strain of FRP bars decreased gradually with an increasing exposure period, which was similar to the phenomenon that appeared in the degradation of tensile strength. Besides, by comparing the degradation trend of the ultimate strain shown in Fig. 11, the decline process of the ultimate strain could be divided into three stages. In the first stage, with an exposure period less than 45 days (Stage I in Fig. 11), the downward of the ultimate strain generally exhibited a slow trend. With the exposure time increasing from 45 to 90 days (Stage II in Fig. 11), a faster decline rate of the ultimate strain appeared in this second stage. In the last stage with an exposure period of 90–180 days (Stage III in Fig. 11),

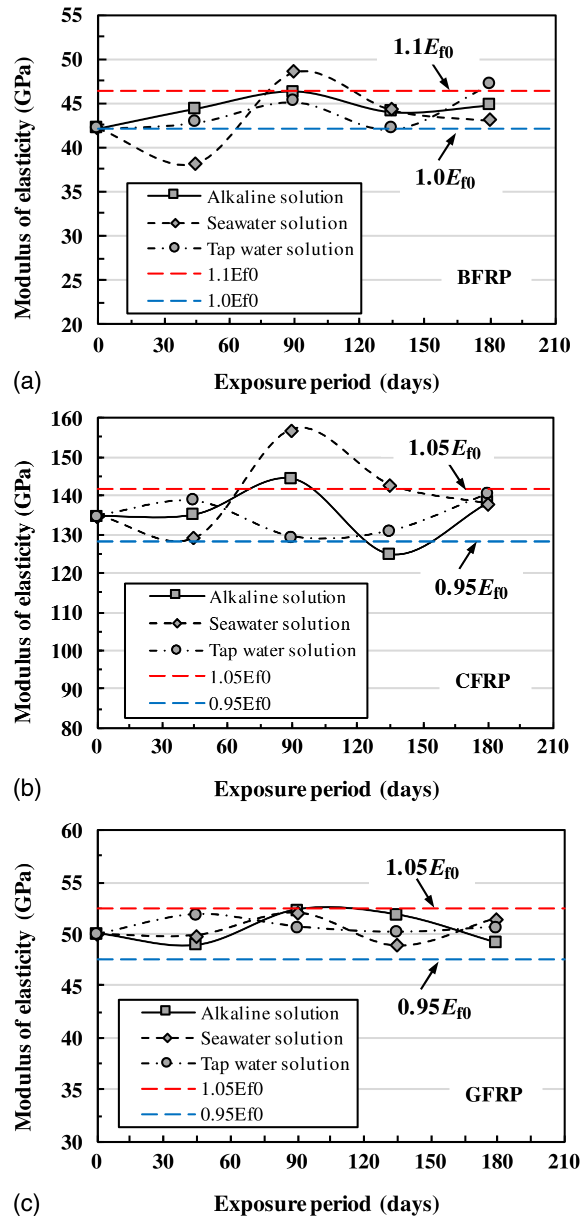

Fig. 10. Relationships between elastic modulus of FRP bars and exposure periods: (a) BFRP bars; (b) CFRP bars; and (c) GFRP bars.

the degradation of ultimate strains became stable, and sometimes a slight increase of the ultimate strain could be observed in this stage.

From previous experimental and analytical results, it can be concluded that for the FRP bars tested in this study, the tensile strengths and their corresponding ultimate strains decreased in the aggressive solutions while no significant change in the modulus of elasticity was observed. Therefore, the influence of the aggressive solution on the bar elastic modulus is negligible and the following discussion will be focused on the degradation of the tensile strength of all tested FRP bars.

Prediction Model of Tensile Strength of FRP Bars

In order to analyze the short-term accelerated test data of FRP bars, the Arrhenius theory has been widely adopted to predict the tensile strength retention of FRP bars by researchers (Chen et al. 2006, 2007; Robert and Benmokrane 2013; Wu et al. 2015;

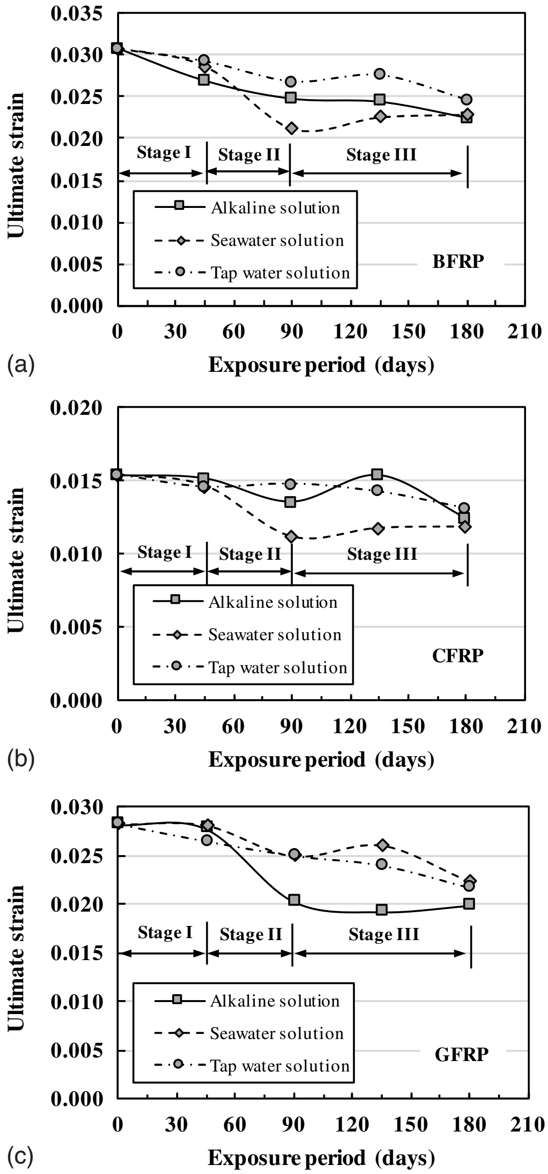

Fig. 11. Relationships between tensile ultimate strain and exposure period: (a) BFRP bars; (b) CFRP bars; and (c) GFRP bars.

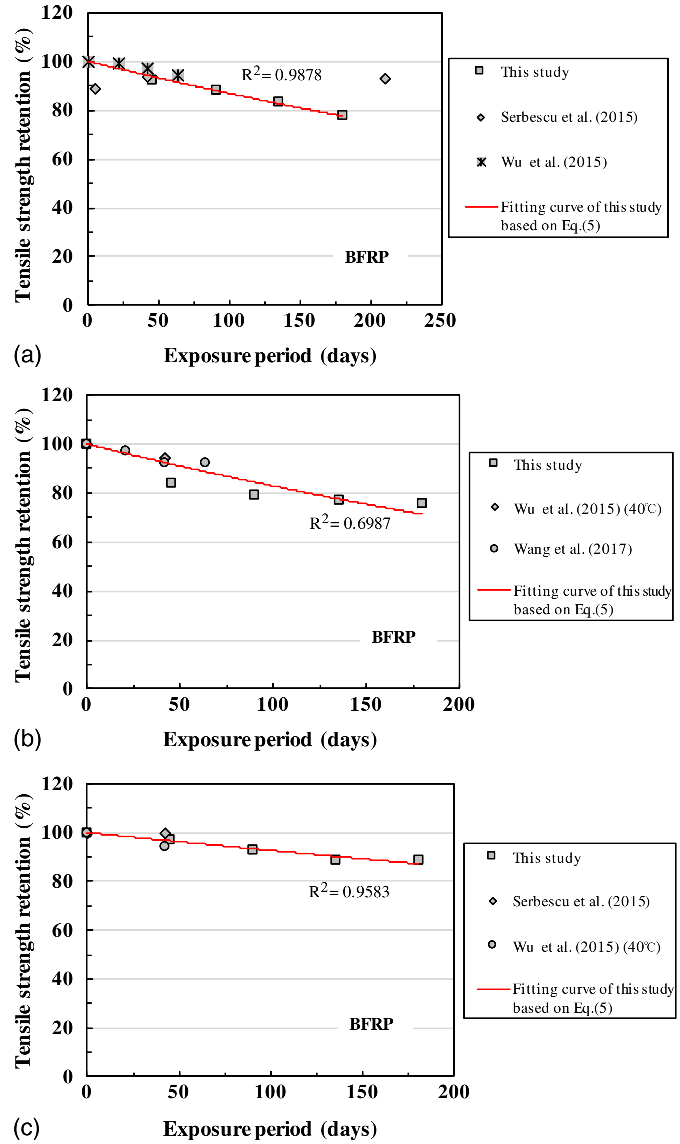

Fig. 12. Fitting results of BFRP bars and comparisons with other literatures: (a) alkaline solution; (b) seawater solution; and (c) tap water solution.

Wang et al. 2017; El-Hassan et al. 2018; Wang et al. 2018). However, it should be noted the basic assumption of this theory is that the single dominant degradation mechanism of material does not change with time and temperature during the exposure, while the degradation rate is accelerated with increasing the temperature (Chen et al. 2006; Wang et al. 2017). Some experimental results (Wu et al. 2015) indicated that the degradation mechanism of FRP composites maybe changed with the exposure period, violating the basic assumption of the Arrhenius theory (Davalos et al. 2012). Although potential drawbacks exist, many predicting models have been proposed based on this theory. Among them, the three following models were most frequently used by previous researchers, as given in Eqs. (4)–(6), respectively

$$\eta(t) = a\log(t) + b \tag{4}$$

$$\eta(t) = 100\exp(-t/\tau) \tag{5}$$

$$\eta(t) = (100 - \eta_{\infty})\exp(-t/\tau) + \eta_{\infty} \tag{6}$$

where $\eta(t)$ = time-dependent tensile strength retention; t = exposure time; a and b = two regression parameters; τ = fitting constant; and $\eta\infty$ = tensile strength retention at exposure infinite period.

For the first model given in Eq. (4), some researchers (Wu et al. 2015; Wang et al. 2017, 2018) indicated that there are some intrinsic limitations due to the fact that this model cannot provide any hypothesis for the degradation mechanism. For the other two models shown in Eqs. (5) and (6), their degradation mechanisms related to FRP bars are assumed to be debonding at the fiber/matrix interface (Wu et al. 2015, 2018), and they have been successfully used to fit the short-term experimental data of tensile strength retention of BFRP bars exposed to two kinds of environmental solutions (Wang et al. 2017, 2018). However, it should be noted that the value of $\eta\infty$ should be determined firstly when the model in Eq. (6) is applied to a fitting analysis and that the model's precision will be obviously influenced by the determination of $\eta\infty$. Therefore, based on the successful applications to BFRP bars (Wu et al. 2015, 2017, 2018) and GFRP bars (Chen et al. 2006; Davalos et al. 2012), only the model

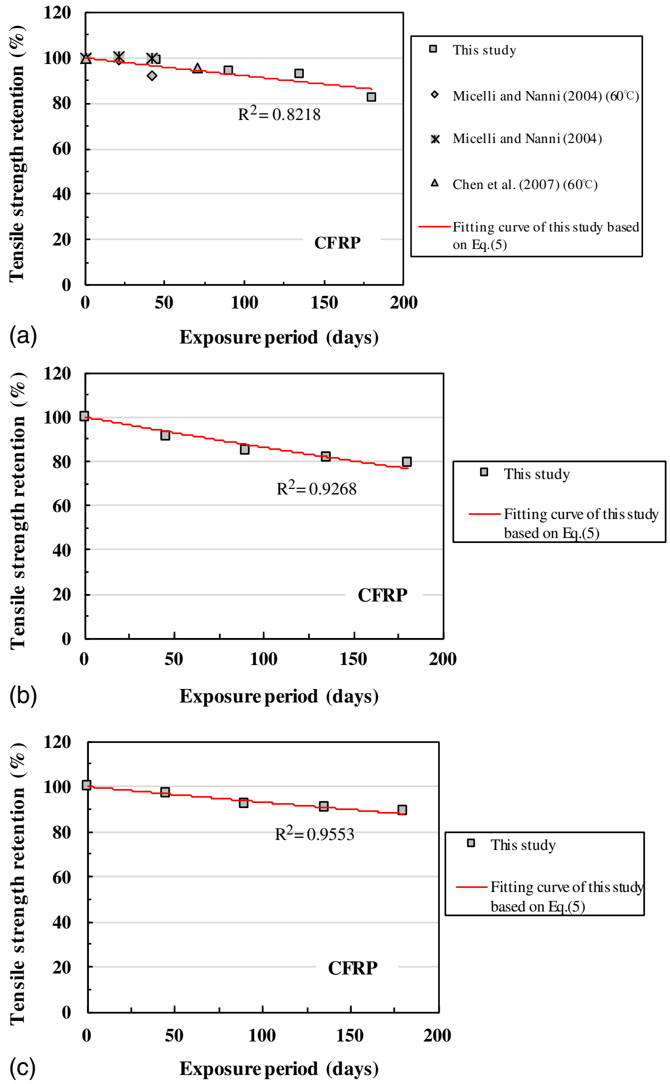

Fig. 13. Fitting results of CFRP bars and comparisons with other literature: (a) alkaline solution; (b) seawater solution; and (c) tap water solution.

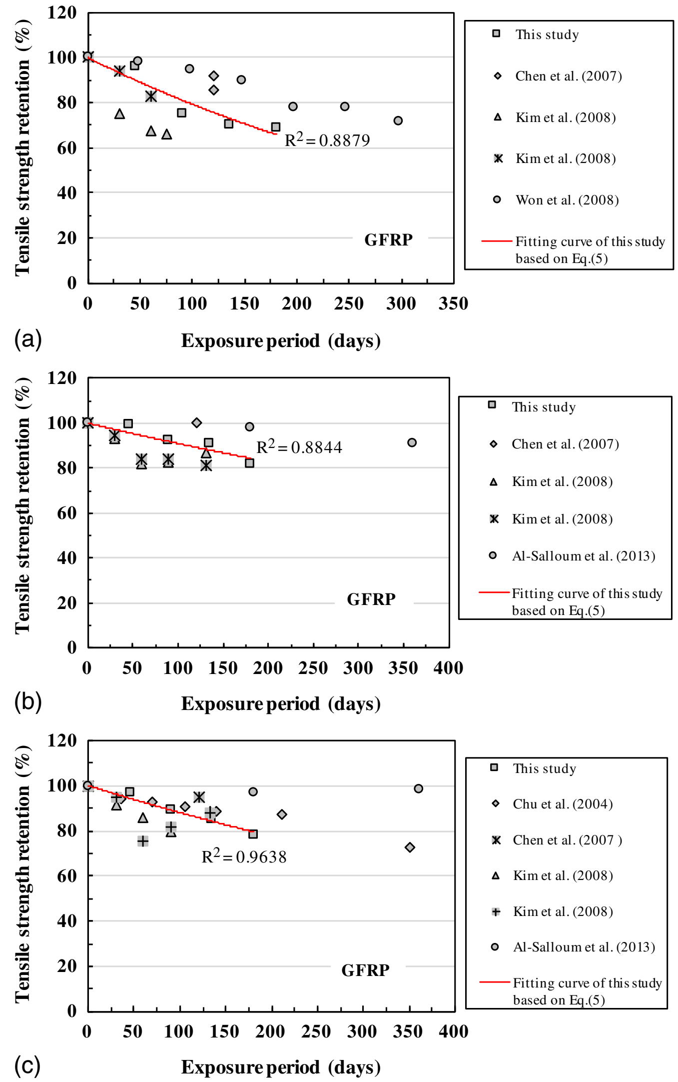

Fig. 14. Fitting results of GFRP bars and comparisons with other literature: (a) alkaline solution; (b) seawater solution; and (c) tap water solution.

given in Eq. (5) will be adopted to fit the experimental data obtained in this study.

Based on the experimental calculations of the tensile strength retention of tested FRP bars, as shown in Figs. 12–14, the fitting analysis was performed with Eq. (5), the obtained τ values, and the determination coefficients ($R2$) for different FRP bars, and exposure solutions were listed in Table 4. It can be found from Table 4 that except when the BFRP bar is immersed in seawater solution, the other obtained values of the determination coefficient $R2$ were in the range of 0.82–0.99, which exceeded the recommended minimum value of 0.80 for acceptability (Ali et al. 2019). Therefore, the preceding fittings based on Eq. (5) could be believed to be basically reasonable for all FRP bars tested in this study. A comparison of the results given in Table 4 showed that for BFRP and CFRP bars, their minimum values of the fitting constant τ were from the seawater solution, while the minimum τ of GFRP bars appeared in the alkaline solution. Furthermore, taking three types of FRP bars and three kinds of aggressive solutions into account together, the τ value obtained from GFRP bars immersed in the alkaline solution was smallest, which meant that the degradation rate of GFRP bars in the alkaline solution was the highest. These findings were consistent with the experimental observations given in Fig. 9.

Specially, for BFRP bars tested in this study, Fig. 12 and Table 4 showed that the τ values were 709.2 and 531.9 for the alkaline solution and seawater solution at around 20°C, respectively. Wu et al. (2015) reported the τ value of 1,412.8 for BFRP bars exposed to the

Table 4. Results of fitting analysis with Eq. (5) for all FRP bars

Bar type	Environmental solution	Fitting results τ	Fitting results R^2
BFRP	Alkaline solution	709.2	0.9878
	Seawater solution	531.9	0.6987
	Tap water solution	1,319.3	0.9583
CFRP	Alkaline solution	1,224.0	0.8218
	Seawater solution	676.1	0.9268
	Tap water solution	1,392.8	0.9553
GFRP	Alkaline solution	434.8	0.8879
	Seawater solution	1,059.3	0.8844
	Tap water solution	786.8	0.9638

alkaline solution at 25°C, and Wang et al. (2017) showed the τ value of 719.2 for BFRP bars in normal seawater and sea sand concrete (N-SWSSC) solution at 32°C. For our GFRP bars, it can be understood from Fig. 14 and Table 4 that the τ value was 434.8 in the alkaline solution. Chen et al. (2006) obtained two values of 256 and 1,667 for two kinds of GFRP bars with different types of E-glass fibers immersed in alkaline solution 1 and 2 (Table 1) at 20°C, respectively. Comparing the previous fitting results of this study and other studies, it should be realized that the degradation rate of the same kind of FRP bar seems to vary in a wide range, which may be attributed to the differences in material properties of FRP bars and aggressive environmental conditions.

In order to testify the reliability of test data obtained in this study, many experimental results tested on the three kinds of FRP bars in other literatures were collected and also plotted in Figs. 12–14 accordingly. Because the solution immersion tests in this study were carried out at room temperature, most of the collected data were also obtained around 20°C. However, due to the lack of comparative data, some experimental results at a higher temperature than 20°C were adopted, and the corresponding temperature was added after the literature.

For BFRP bars, the comparison shown in Fig. 12(a) indicated that the our tensile strength retentions in the alkaline solution were equivalent to the results given in Serbescu et al. (2015) and Wu et al. (2015). It can be found from Figs. 12(b and c) that the similar results of strength retentions in seawater and tap water solutions were obtained in this study and other literatures (Wu et al. 2015; Serbescu et al. 2015; Wang et al. 2017). In particularly, the determination coefficient obtained from our specimens in the seawater solution was a low value of 0.6987 [Fig. 12(b)]; however, the experimental data proposed by Wang et al. (2017) were well consistent with the fitting curve.

For CFPR bars exposed to the alkaline solution, it can be observed from Fig. 13(a) that the degradation rate of the tensile strength seemed to be similar, comparing the results reported in this study and other investigations (Micelli and Nanni 2004; Chen et al. 2006). Besides, due to the lack of experimental data, no comparison had been added in Fig. 13(b) and c for CFRP bars immersed in seawater and tap water solutions.

For GFRP bars exposed to aggressive solutions, many experimental investigations related to the degradation of tensile strength have been completed in the recent decade. The results given in Fig. 14(a) showed a great discrepancy in the tensile-strength retention of GFRP bars immersed in an alkaline solution. It can be noticed that the outcomes from this study fell in the middle of these data. The comparisons shown in Figs. 14(b and c) indicated that the difference in strength degradation of GFRP bars exposed to seawater and tap water solutions was not as obvious as it appeared in the alkaline solution. The corresponding degradation trends proposed in this study agreed well with the results reported in Kim et al. (2008) and Chu et al (2004).

Conclusions

The following observations and conclusions can be drawn from this study:

- After 180 days of exposure to tap water, seawater, and alkaline solutions, the appearance changes of GFRP bars were more obvious than those of BFRP and CFRP bars. By comparing, it can be basically concluded from the surface appearance that the ability of chemical resistance of CFRP bars was best among the three kinds of FRP bars tested in this study. This phenomenon may be attributed to the different surface treatments and moisture absorption contents of tested FRP bars.
- In the tensile test, most of the tested FRP bars showed a similar failure mode in which the bar failed through fiber rupture regardless of the aggressive solution and exposure period. This failure was accompanied by a longitudinal delamination of fractured fibers, and the fiber dispersion of GFRP bars was greater than those of the BFRP and CFRP bars.
- The load-displacement curves obtained from the tensile test indicated that all unconditioned and conditioned FRP bars presented an approximately linear behavior up to failure. The tensile strength of the conditioned FRP bars degenerated obviously with an increasing exposure period, while no significant degradation of the elastic modulus of the three kinds of FRP bars had been found after exposure. Comparisons of test results showed that the CFRP bars had a better resistance ability to chemical attack than the BFRP and GFRP bars. Due to the differences in moisture absorption and susceptibility to immersion solutions, GFRP bars exposed to alkaline solution showed the most serious degradation of tensile strength, while the seawater and alkaline solutions caused a significant damage on strength loss of BFRP bars.
- With an increasing exposure time, the decline process of ultimate strains of conditioned FRP bars could be divided into three stages: a slow downward with the exposure period less than 45 days, a fast decline appeared between the periods of 45 and 90 days, and a stable process within the exposure period of 90–180 days,
- Based on the Arrhenius theory, the predicting model of tensile-strengthen retention shown in Eq. (5) was adopted to fit the experimental data obtained in this study. Comparing the fitting results of this study and other studies, it can be concluded that our experimental data could be well fitted by Eq. (5) with determination coefficients in the range of 0.82–0.99 and that they also agreed well with most of the collected data. However, it should be realized that the tensile strength degradation of the same kind of FRP bars, such as GFRP bars, seemed to vary in a wide range, which might be attributed to the differences in material properties of FRP bars and aggressive environmental conditions.

Data Availability Statement

The following data, models, or code generated or used during the study are available from the corresponding author by request: the data of the moisture uptake of tested FRP bars; the data of the heat flow first derivative of tested FRP bars in the DSC analysis; and the data of the load-displacement curves of tested FRP bars.

Acknowledgments

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